

Weeks 2-10

Grades 4-5

Investigate all cases, prove an obstruction, or explain an optimum.

Grade bands are entry points. Begin with objects, then try to explain what happens.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Keep facilitator solutions separate from student pages.

Week and investigation	PDF pages
02 Switches and lamps	2-5
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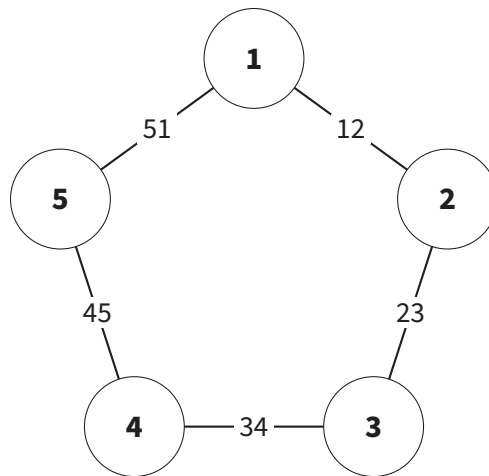
Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

Revised September 27, 2026. Unpiloted revisions; record actual use after teaching.

Problem 1: Start with five two-sided counters OFF. Pressing a line flips both end lamps, including ON to OFF. Write 12 for pressing the line from 1 to 2. Try two moves and undo them.



Problem 2: Start all OFF for each target. Make only the listed lamps ON. Find two different move lists, then circle your shorter list.

ON lamps	First list	Another list
1 and 2		
1 and 3		
1, 2, 3, 4		

Choose a target. Test whether one press can make it. Record what happened.

Problem 3: Start all OFF for each list. Try 12,23; then 23,12; then 12,23,12,12. Draw the final lamps for each. Cross out any pair of repeated presses that you can remove without changing the final picture.

Problem 4: A reduced list uses each edge at most once. For each reduced list below, make its target. Reset; press exactly the edges missing from that list. Compare the two final pictures and count the presses. The five edges are 12,23,34,45,51.

First list	Missing edges	Final ON lamps	Counts
12			
12,23			
12,34			
12,23,34			

Problem 5: Choose a new edge set and its missing edges. Test both. Record the two press counts and whether the final targets agree.

Problem 6: Use the five-lamp mat. To leave only lamps 1 and 3 ON, first choose whether to press 51. Lamp 1 then tells you whether to press 12. Keep going around the ring. Fill yes or no, test the full list, and check lamp 5 last.

First choice	51	12	23	34	45	Lamp 5
No	no					
Yes	yes					

Problem 7: Repeat the forced choices for only lamps 1,2,3,4 ON. Compare your two lists with your experiments on page 1.

First choice	51	12	23	34	45	Lamp 5
No	no					
Yes	yes					

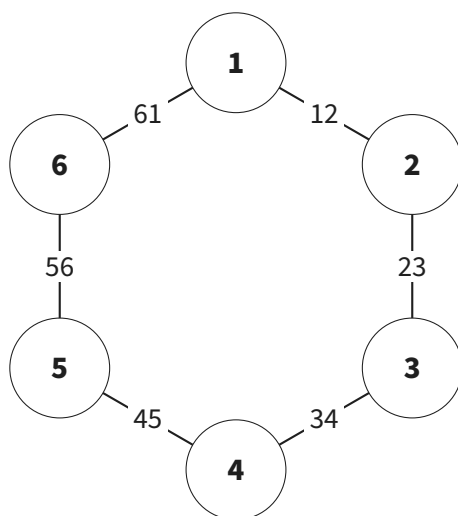
Problem 8: Now try to leave only lamp 1 ON. Fill both rows. Put a cross beside any row whose final lamp does not match the target.

First choice	51	12	23	34	45	Lamp 5
No	no					
Yes	yes					

Record a rule you notice about the two lists or their targets.

Problem 9: On the five-lamp ring, choose three new targets with an even number of ON lamps. Find the shortest reduced list for each. Record both complementary lists and their lengths. Try to find a target needing more than two presses.

Problem 10: Use six counters on this ring. Start all OFF. Make only lamps 1 and 4 ON in two different ways. Count the presses. Then repeat for only 1 and 3 ON. Compare the two lengths for each target.



Move lists, lengths, and a comparison with the five-lamp ring:

Problem 1: Use five letter cards. One turn changes every letter using the chosen key, keeping message positions fixed. Start each experiment with ABCDE. Repeat the same key and record each message until the whole row first returns. Circle that turn.

Input	A	B	C	D	E
Key R	B	C	D	E	A
Key S	B	C	A	E	D

Turn	R message	S message
0	ABCDE	ABCDE
1		
2		
3		
4		
5		
6		

Problem 2: For each key, follow A and draw its arrows until it comes back. Then start at a letter not yet drawn. Draw every loop, including any letter that stays put. Compare the two keys' loop sizes and return times.

Problem 3: Build three more keys from five letter cards: loops of sizes 4+1, then 2+2+1, then 3+1+1. Draw each key and test when all letters first return.

Loop sizes	My arrows	Return
4 + 1		
2 + 2 + 1		
3 + 1 + 1		

Problem 4: Return to key S, with loops of sizes 3 and 2. Move a counter around each loop, one arrow per turn. In each row below, circle every turn when that counter is home. Box the first positive turn circled in both rows.

Loop	1	2	3	4	5	6	7	8	9	10	11	12
Size 3	1	2	3	4	5	6	7	8	9	10	11	12
Size 2	1	2	3	4	5	6	7	8	9	10	11	12

Problem 5: Invent another five-letter key. Predict its return from its loops, then test. Record your prediction and result.

Problem 6: Arrange five letter cards into loops. Organize different arrangements by their largest loop: 5, then 4, then 3, then 2, then 1. For each largest loop, split the remaining cards in every way you can. Record sizes and the first common return. Changing letter names does not create a new list of sizes.

Loop sizes	First common return

Problem 7: Choose the arrangement with the longest return. Write a key that has those loops. Test it with counters and mark its return turns. Compare with your earlier keys.

Problem 8: Key P swaps A and B. Key Q swaps B and C. Both leave all other letters alone. Start with ABCDE each time. Apply the first key, record the whole row, then apply the second key to that row.

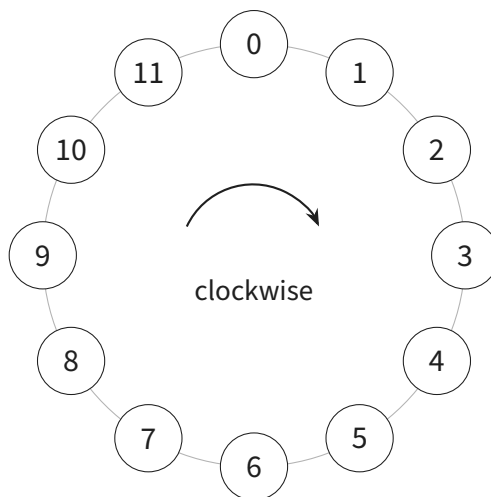
Order	After first key	After second key
P then Q		
Q then P		

Problem 9: Treat P then Q as one turn. Repeat that whole turn on ABCDE until the row first returns. Draw the loops of this combined key.

Problem 10: Change Q to a new key T that swaps D and E and fixes A,B,C. Test P then T and T then P. Compare this pair with Problem 8.

Problem 11: Start with ABCDE and apply P then Q. Now find an order of these same two keys that undoes the result. Record the full check.

Problem 1: A shift $+k$ moves k places clockwise, wrapping after 11. Try $+4$ from 0, then from 9. Record both landings. Find where inputs 2 and 8 go under the same shift.



Problem 2: A secret shift takes 9 to 1. Use the ring to recover it. Test the additional clue 2 to 6. Decode outputs 1,4,10 by undoing that shift.

Problem 3: Repeat $+4$ from 0 until it first returns. Write the whole route. Start at a position not yet used and repeat. Continue until every position is in a recorded loop.

Problem 4: Use the twelve-place ring. Start at 0 and repeat +8 until you return. Find its other loops. Compare the loop members and their order with +4.

Problem 5: Try +3, then +6, then +5. For each rule, find the route from 0 and all remaining loops. Record the first return time and the number of loops.

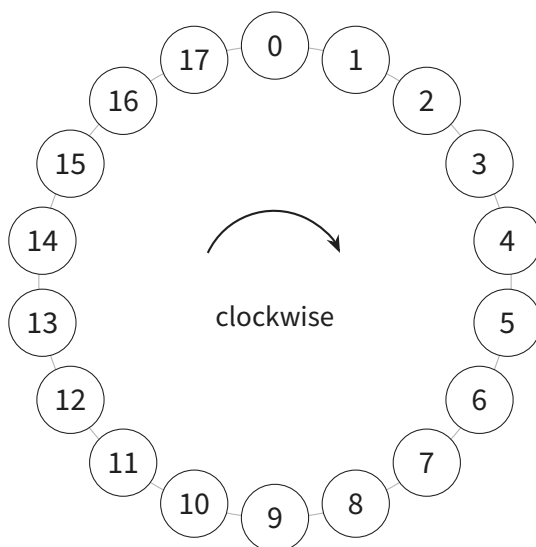
Shift	Routes	Return	Count
+3			
+6			
+5			

Problem 6: Compare these shifts with +4 and +8. Group shifts with the same return time. Record a prediction connecting the step size, ring size, and number of loops.

Problem 7: Test the remaining shifts on the twelve-place ring. Record first positive return times and number of loops. Use your earlier results where you have them. For +0, a turn leaves the counter in place.

Shift	Return	Loops	Shift	Return	Loops
0			6		
1			7		
2			8		
3			9		
4			10		
5			11		

Problem 8: On this eighteen-place ring, predict the first return for +6 and +5. Test both and record their routes from 0. Compare with your prediction.



Problem 9: Each +4 turn on twelve places moves four spaces. Fill the distance table and circle the first positive total that is a whole number of laps. Repeat for +8. Compare this distance record with the routes you drew.

Turns	+4 distance	Whole laps?	+8 distance	Whole laps?
1				
2				
3				
4				

Problem 10: Find a shift that undoes +4 on twelve places. Test it from 0,1,5, and 11. Record the four out-and-back trips.

Problem 11: Start at 0, then 2, then 9. Compare +4 followed by +7 with +7 followed by +4. Record both final landings for each start. Find one shift that does the same job.

Problem 12: Choose another pair of shifts on twelve places. Test both orders from two starts. Record your combined shift and a rule suggested by the experiments.

Problem 1: Build these rows with heights 1, 2, 3, and 4. Look from both ends. Count visible towers; a taller tower hides any shorter tower behind it.

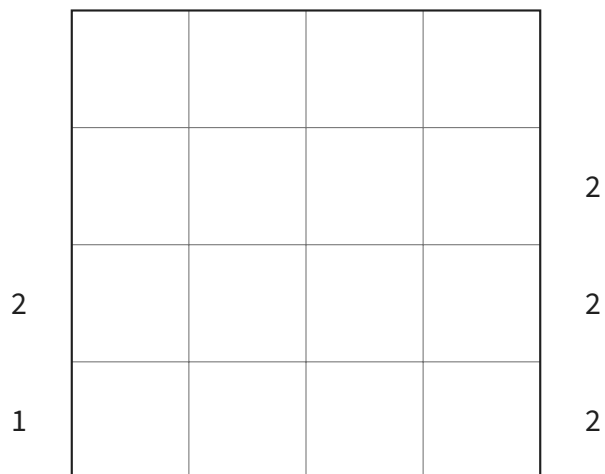
Order, left to right	Seen left	Seen right
1 4 2 3		
2 1 4 3		
4 1 2 3		
4 2 1 3		

Problem 2: Build a city using heights 1-4 once in every row and once in every column. Each outside number counts towers seen from that end. Solve the city below and record each tower height.

	4				
4					
					2
2					2
1					2

One tower I placed and the clue I used:

Problem 3: Remove the top clue 4 and the left clue 4. Rebuild using only the five remaining clues below. Keep a record of any different city you find.



Problem 4: Use four towers to test bottom rows before filling the city. Record every bottom row that shows one tower from the left and two from the right. Keep both possibilities available.

Possible bottom rows, left to right:

Problem 5: Now build rows showing two towers from each end. Record all the different rows you find. These are candidates for row 3, counting rows from the top.

Possible row 3 orders:

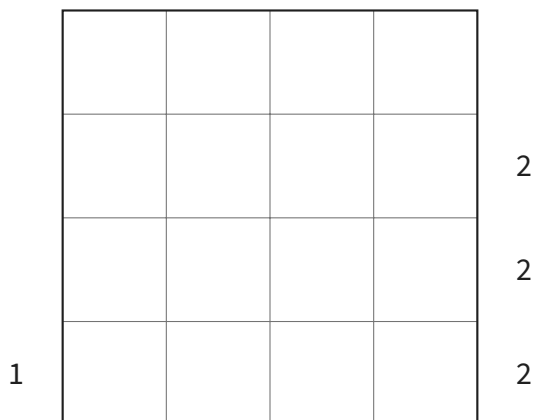
Problem 6: Pair each bottom row from Problem 4 with each row 3 from Problem 5. Reject a pair if a column repeats a height. Record the pairs that survive. For each pair, try every row 2 that avoids column repeats and shows two towers from the right. Write all the row 2 orders that work, or write “none.”

Bottom row	Row 3	Row 2 orders that work

Problem 7: For each case that survives Problem 6, fill the top row with the heights missing from its columns. Record the completed city. Check all five clues again.

				2
2				2
1				2

Problem 8: Start with the five-clue city from Problem 7. Cover the left clue on row 3. Try to make a different city that keeps the other four clues. Record it, then restore the covered clue.



Problem 9: Check the alternatives below one at a time. L3 means the left clue on row 3; R2 means the right clue on row 2. In each city, a slash begins the next row, from top to bottom. Check every row and column and the four retained clues. Record the actual view at the covered clue. Restore it before the next check.

Cover	Alternative city	Actual view
L3	1234 / 3412 / 2341 / 4123	
L4	1234 / 4123 / 3412 / 2341	
R2	1324 / 2431 / 3142 / 4213	
R3	1324 / 3142 / 2431 / 4213	
R4	1234 / 2143 / 3412 / 4321	

Problem 10: Remove every old clue. Use only left clues 4, 3, and 2 on rows 1, 2, and 3. Build a city with these clues and record its tower heights.

4				
3				
2				

Problem 11: List the top rows that work with its clue. Then list the row 2 orders that fit its clue and avoid repeats under the top row. Continue with row 3. Fill row 4 with each column's missing height.

Row	Orders that fit all earlier rows and this row's clue
1	
2	
3	
4	

Problem 12: Compare this city with your five-clue city. Record how many clues each puzzle used and whether the completed cities agree.

My comparison:

Problem 1: Use R and B counters; repeats are allowed. A test scores one point for each matching position. Say only the total score. Use four positions. Hide a secret and make tests until you can identify it. Keep every test and score visible; add rows on scrap paper if needed. Naming a determined secret costs no extra test. Reveal and check your record.

Test	Code tested	Score
1		
2		
3		
4		
5		

Problem 2: Build each visible secret below. Test RRRR, then BRRR. Only the first test position changes. Record the scores and their difference.

Secret	RRRR score	BRRR score	Change
RBRB			
BRBR			
RRRR			
BBBB			

Problem 3: Keep a visible secret while you compare RRRR with each of BRRR, RBRR, and RRBR. Each comparison changes one position from the RRRR starting test. Record the scores for all four secrets.

Secret	RRRR	BRRR	RBRR	RRBR
RBRB				
BRBR				
RBBB				
RRRR				

Problem 4: Try those four tests on a hidden secret. Use each change from the RRRR score to find positions 1, 2, and 3. Count the Rs you have found. Subtract that count from the RRRR score to decide position 4. Record your answer, then reveal and check.

Test	Code tested	Score
1		
2		
3		
4		

My four positions and the baseline R count:

Problem 5: The first test RRRR scored 2. Build the six possible secrets below. For each proposed second test, record its score against each secret. In each row, circle the largest group of equal scores.

Test	BBRR	BRBR	BRRB	RBBR	RBRB	RRBB
BRRR						
BBRR						
BBBR						
BBBB						

Problem 6: For the second test BRRR, keep the secrets that would score 3. For the second test BBRR, keep the secrets that would score 2. Copy each group below. Try one final test against every secret in its group; record its scores.

BRRR answered 3: candidates, final test, and scores:

BBRR answered 2: candidates, final test, and scores:

Problem 7: Keep the three candidates BBRR, BRBR, and BRRB. Try the final tests below. Circle repeated scores within each row. Add a test of your own.

Final test	BBRR	BRBR	BRRB
RRRR			
RBRR			
RBBR			

Problem 8: Keep the four candidates BRBR, BRRB, RBBR, and RBRB. Try these final tests and circle repeated scores. Add a test of your own.

Final test	BRBR	BRRB	RBBR	RBRB
RRRR				
BRRR				
BBRR				

Problem 9: Take the first test BRBR and a secret of your choice. Swap R with B at positions 1 and 3 in both words, turning the test into RRRR. Record the two secret words and both scores. Repeat with a different secret.

Original secret	Score for BRBR	Changed secret	Score for RRRR

Problem 1: Two players alternate taking 1, 3, or 4 counters. Taking 2 is not allowed. Never take more than remain. Taking the last counter wins. Play from 8, 10, and 14. Record your moves. Swap who starts for a second attempt at a pile that surprised you.

Starting pile	First move	Next moves, in order	Winner: first or second
8			
10			
14			

Problem 2: W means the next player can force a win; L means every legal move gives the opponent a winning strategy. Zero is L. Build small piles and work upward: mark W if a legal move reaches L, and L if every legal move reaches W. Record the next piles as well as the labels.

Pile	Legal next piles	W/L
0	none	L
1		
2		
3		
4		
5		
6		
7		

Problem 3: Continue using moves 1, 3, and 4. Fill the chart in order. In each cell write L, or write W and a winning move: W1 means W with winning move 1. Check all legal next piles before marking L. Compare the rows after filling at least two of them.

Pile	0	1	2	3	4	5	6
Label/move							
Pile	7	8	9	10	11	12	13
Label/move							
Pile	14	15	16	17	18	19	20
Label/move							
Pile	21	22	23	24	25	26	27
Label/move							

Problem 4: Compare these four piles. Record every legal next pile and its label. Circle the starting piles that have no move to L.

Pile	Take 1: next pile/label	Take 3: next pile/label	Take 4: next pile/label
14			
16			
15			
17			

The repeating pattern I see:

Problem 5: Arrange 14, 16, 15, and 17 counters in whole groups of 7 with a smaller leftover group. Record the leftovers. Match each pile to its label in Problem 4.

Pile	Whole groups of 7	Leftover counters	W/L
14			
16			
15			
17			

Problem 6: Use your chart to choose a winning move for each leftover count marked W. Record the leftover count after the move. For counts marked L, list every leftover count an allowed move can reach; use piles 7 and 9 so that all three moves are possible.

Leftover count	W/L	Move to L, or all moves from L	New leftover count(s)
0			
1			
2			
3			
4			
5			
6			

Problem 7: Build 24 counters. Choose a first move using your table, then play. After each of your turns record the leftover count when grouping the remaining pile in sevens.

First move and later leftover counts:

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Problem 8: Change the allowed moves to 1, 3, and 5. Play from 6 and from 7, then swap who starts. Build smaller piles to check a proposed strategy. Record which piles you want to hand the other player.

My games and the piles I want to leave:

Problem 9: Return to moves 1, 3, and 4. Copy the labels for piles 0–3 into the first row below and 7–10 into the second. These are four-label windows. To find the next label, inspect the first, second, and fourth cells: mark the next W if any is L, and L if all three are W. Record the next label, then slide the window one cell right.

Window starts at	Four labels	Next label	Window after sliding
0			
7			
1			
8			

Problem 10: Find another matching pair of four-label windows in your chart. Copy both windows, extend them by two labels, and compare the results.

My matching windows and their next two labels:

Problem 1: Build three piles. Take turns removing one or more counters from exactly one pile. Whoever takes the last counter wins. Play (1,2,3), then (1,2,4). Change who starts and play each again. Use the spaces for your piles.

Start	Who won?	One first move and reply
(1,2,3)		
(1,2,3)		
(1,2,4)		
(1,2,4)		

Problem 2: Replay one game. Choose a different first move and record the reply. Circle a start you would prefer to give your opponent.

Problem 3: Test these starts with counters. For each, try to leave two equal piles and one empty pile in a single move. Write a move if you find one, or record what you tried.

Start	Total	Move or attempts
(1,1,1)		
(1,1,2)		
(1,2,3)		
(2,4,6)		

Problem 4: Rebuild (1,2,3) after each trial. Your partner chooses each first move in this table. Find a reply leaving two equal piles and one empty pile; write the result.

Partner leaves	My reply leaves
(0,2,3)	
(1,0,3)	
(1,1,3)	
(1,2,0)	
(1,2,1)	
(1,2,2)	

Problem 5: Group each pile into bundles of 4, 2, and 1, using each size at most once. Record 1 if you use that bundle, 0 if you do not. The first row shows 5 counters grouped as $4+1$. Make and record the other piles with counters.

Pile	4	2	1	Bundle sum
5	1	0	1	$4+1$
1				
2				
3				
4				
6				
7				

Problem 6: Represent these two games using the same bundle records. Count the 1s in each column. Circle every odd count. Call a game balanced if all three counts are even; zero is even.

Pile	4	2	1	Pile	4	2	1
1				1			
2				2			
3				4			
Count				Count			

Problem 7: For (1,2,4), reduce just one pile so all column counts become even. Make that move with ordinary counters. Record the new piles.

Problem 8: For each start, write each pile’s 4/2/1 record. If all column counts are even, write “balanced.” Otherwise shrink one pile to make them even and record the new piles. Make each move with counters.

Start / pile	4	2	1	Move or balanced
2				
4				
6				
3				
4				
5				
1				
5				
7				
3				
5				
6				

Problem 9: Choose one balanced start above. Let your partner make a move. Record the new piles, then find and play a move that restores balance. Repeat with a different partner move.

Partner leaves	My reply leaves

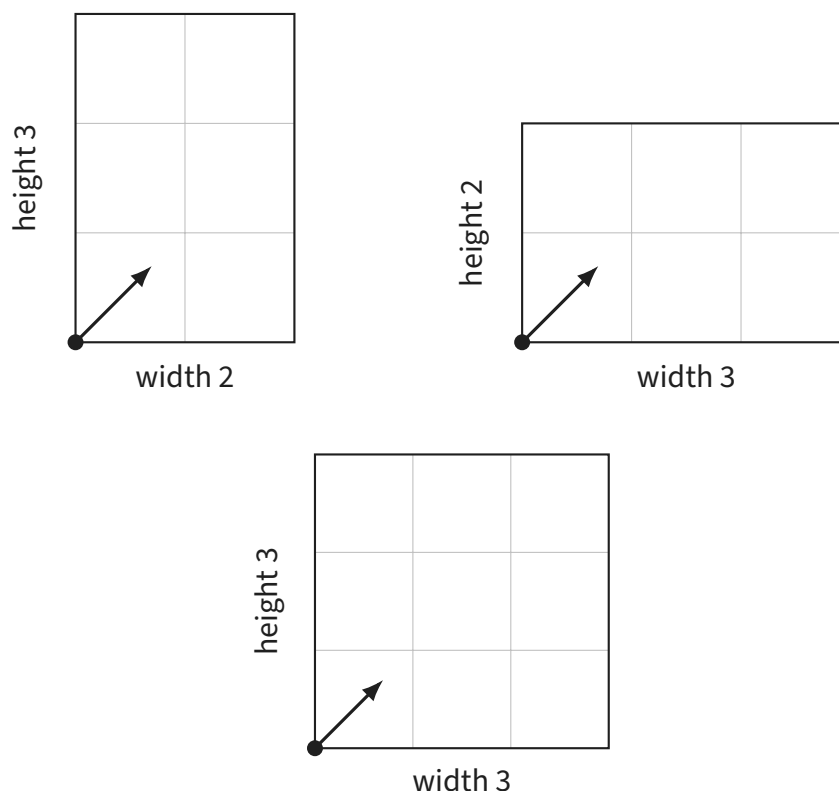
Problem 10: Try this way to repair odd columns. Find the largest bundle column with an odd count. Choose a pile using that bundle. Keep its larger bundles, remove this bundle, and choose its smaller bundles to make every column even. Try (4,6,7), then (7,10,12). Record the starting bundles and the replacement pile. Check that it is smaller, then make the move with counters.

Pile	8	4	2	1	Replacement pile
4					
6					
7					
7					
10					
12					

Problem 11: Choose a third pile that balances (2,5,____), then (3,6,____). Build both games and check their columns.

Problem 12: Choose two new piles and make a third that balances them. Give the game to a partner, then restore balance after their move. Record your start and one pair of moves.

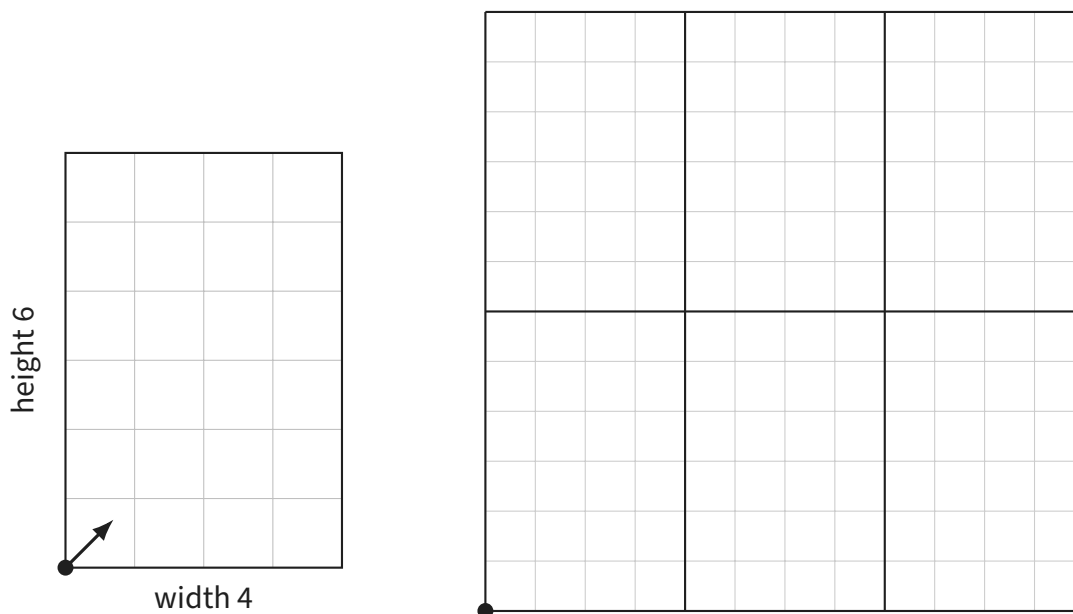
Problem 1: Start at each black dot with slope 1: one unit across for each unit up. Bounce at walls, reversing only the direction blocked by the wall. Stop at the first corner. Trace each path, mark wall bounces with dots, and circle the endpoint. Crossings of old path lines are not walls.



Problem 2: Record the ending corner and number of bounces for each table. Do not count the launch or final corner as bounces. Compare the two non-square tables.

Width $\times$ height	Corner	Bounces
2×3		
3×2		
3×3		

Problem 3: Trace the 4×6 table on the left. On the right are mirror copies of the same room. Draw a straight diagonal from the black dot to the first intersection of a vertical and horizontal room wall. Number the wall crossings before that intersection and match them to bounces on the left.



Problem 4: List the horizontal distances where the straight path crosses a vertical room wall. List the vertical distances where it crosses a horizontal room wall. Circle the first number appearing in both lists. The two distances agree along this diagonal.

Problem 5: Use your circled distance to count whole room widths and heights. Move your finger back and forth across the original width for each whole width, then up and down for each whole height. Record the side reached in each direction and the resulting corner.

Problem 6: Use wall-distance lists to find the first common distance L for each table. Divide it by the width to count whole widths, and by the height to count whole heights. Record the folded corner. Odd whole-width counts end right; even counts end left. Odd whole-height counts end top; even counts end bottom.

Table	L	Widths	Heights	Corner
4×6				
6×9				
6×10				
8×14				

Problem 7: Compare 4×6 with 6×9 . Draw or record how the table and unfolded distances changed. Then compare 4×6 with 6×10 , whose sides are also both even. Circle the two tables with different ending corners.

Problem 8: On the 4×6 copied rooms, extend your imaginary diagonal to distance 24. It would cross 6 whole widths and 4 whole heights. Record the corner those counts predict, then mark the earlier corner where this game actually stops.

Problem 9: For each table, list the distances of vertical and horizontal wall crossings before its first corner. Count all these crossings. Compare your counts with your earlier traced paths.

Table	Vertical walls	Horizontal walls	Bounces
2×3			
3×3			
4×6			
6×10			

Problem 10: Choose a table from Problem 9. Draw its copied rooms or record both wall lists. Circle its final corner separately. Count how many vertical crossings are left when the final corner is removed; then count horizontal crossings. Write a rule using whole-width count m and whole-height count n .

Problem 11: Make a scaled copy of your chosen rectangle by doubling both sides. Predict its corner and bounce count. Check using wall distances and record your result.

Problem 12: Try these three rectangles. Trace their paths and record corners and bounce counts. All have width plus height equal to 6. Circle those reaching top-right after exactly four bounces.

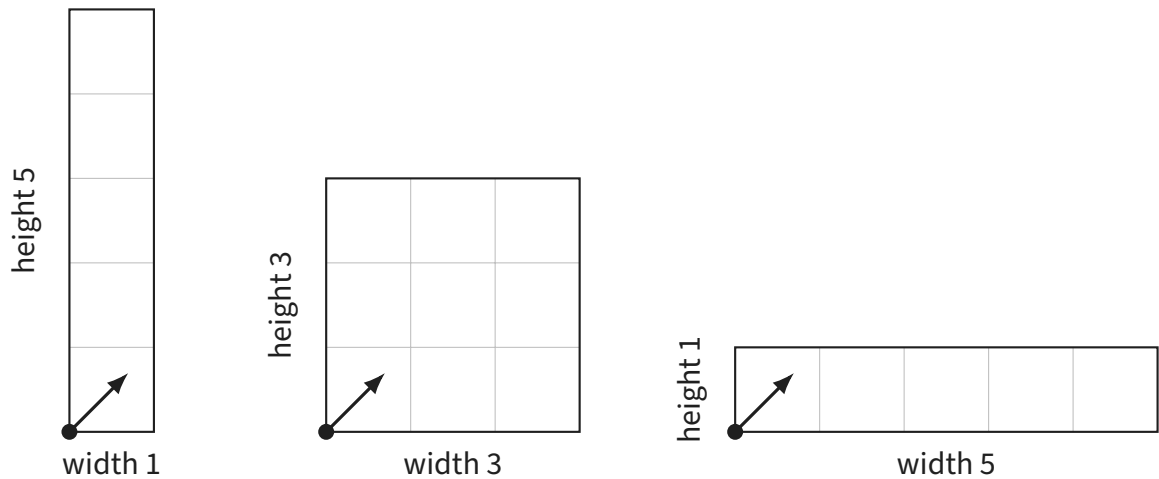
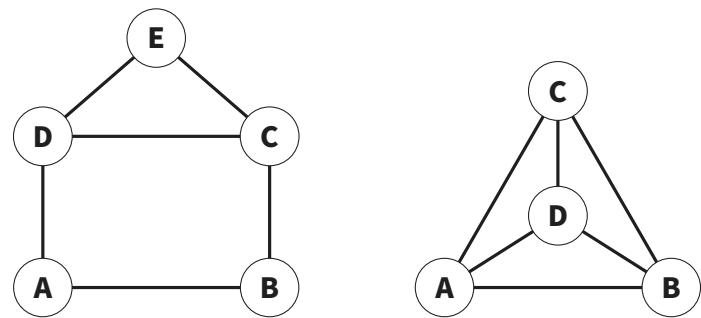


Table	Corner	Bounces
1×5		
3×3		
5×1		

Problem 13: Make a different-size rectangle ending top-right after four bounces. Record its size, first common distance, and whole-room counts.

Problem 14: Try to make a table ending top-right after three bounces. Use the whole-room counts to list your candidates. Record which corner each candidate reaches, or which earlier corner stops the path.

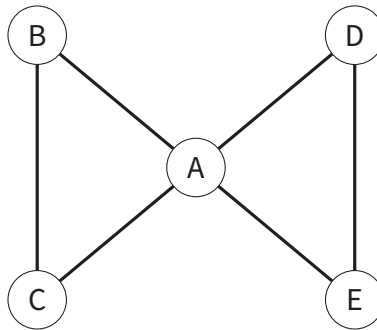
Problem 1: Try to use every line once without jumping. Revisiting a dot is allowed. Mark used lines with counters. Try the house, then the triangle with center D. Choose two different starting dots on each. Record a complete route, or your attempted route and unused lines.



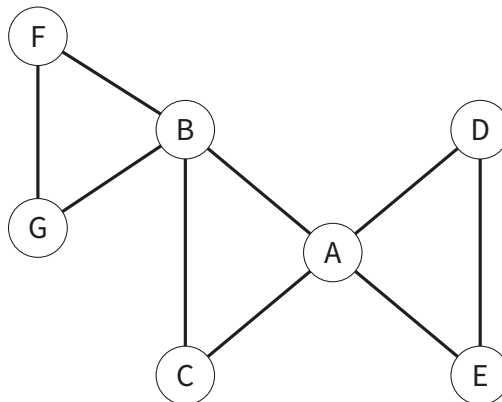
Map	Start	Route	Unused lines
House			
House			
Triangle			
Triangle			

Problem 2: Count each dot’s incident lines, called its degree. Circle the odd-degree dots. Choose your complete house route and pair the arrival/departure lines at each internal visit. Record the dots with one unpaired line.

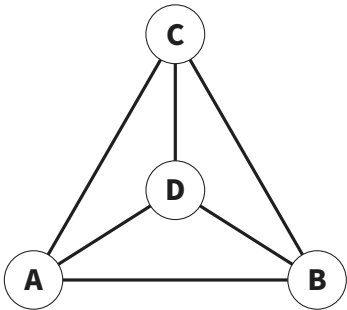
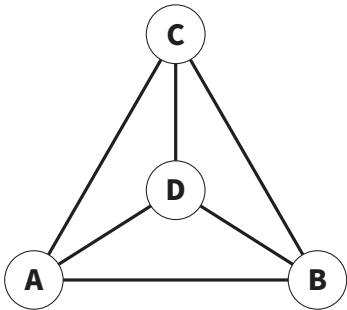
Problem 3: Walk A–B–C–A and mark those roads used. Walk the remaining loop A–D–E–A. Write each loop on a paper strip. Join the strips at A to make one route using all six roads. Record and trace the joined route.



Problem 4: The map below adds a loop at B. Start with your route from Problem 3, then insert B–F–G–B at a visit to B. Record the joined route and check every road is used once. Count the degree at B before and after the new loop.



Problem 5: Now travel every road at least once and return to A. Roads may be repeated; each traversal costs 1 step. Find a route on the left map and record its visited dots. Try a different route on the right map with fewer steps, or match your best count. Mark each repeated road with an extra pencil line.



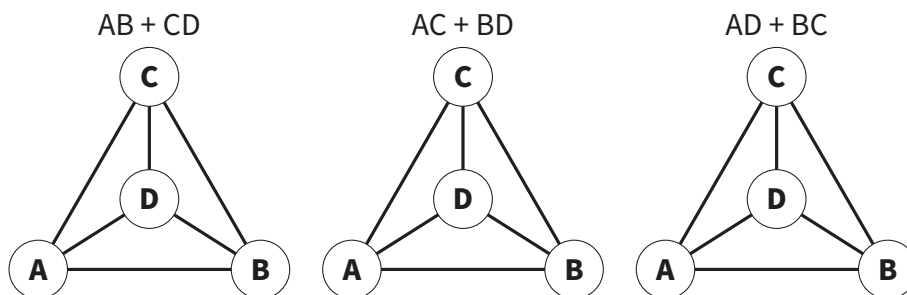
Trial	Route	Steps
1		
2		

Problem 6: Count how many times your best route uses each road. Record the original six crossings separately from the repeats. At each dot, count all arrival and departure crossings, including repetitions. Circle any odd total.

Road	AB	AC	AD	BC	BD	CD
Times used						

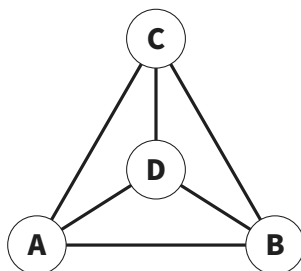
Problem 7: Add one extra copy of a road to a clean map. Count how many odd degrees change to even. Then add one more road copy that makes all degrees even. Record the two roads.

Problem 8: Pair the four odd dots in each way shown. Draw a second copy of each named road on its map. Then find a closed route using every original and added road once. Record its length.

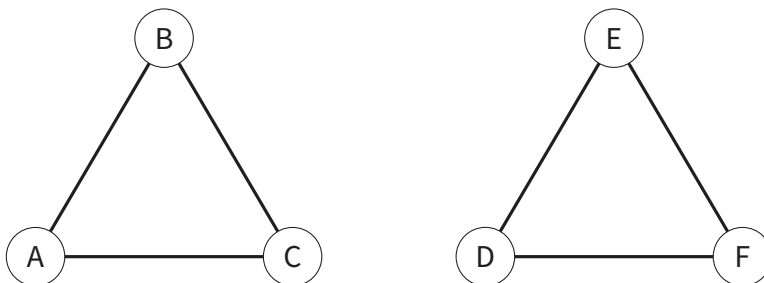


Pairing	Closed route	Length
AB + CD		
AC + BD		
AD + BC		

Problem 9: Now the truck may finish at a different dot. Add just one road copy to the original map below, then find a route using every original and added road once. Record its endpoints and length.



Problem 10: Add three roads between these two triangles so the map is connected and all six dots have odd degree. Curve roads around other dots; a crossing without a dot is not a junction. Count the degrees. Then choose three roads to repeat so every degree becomes even. Record a closed delivery route and its length.



Problem 11: This map also has six odd dots. Start at A, visit every road, and return to A. Mark each repeated road and record your route. Compare how many extra crossings the two maps need.

